Oumaima Chater

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Data Scientist | Intelligence Artificielle | Python, GCP, Machine Learning

Technical Skills

Data Science & Machine Learning: Modèles prédictifs, statistiques, algorithmes ML, GenAI, LLMs, XGBoost

Gestion de Projet Data: Cadrage des besoins métiers, mesure du ROI (A/B testing), data visualisation

Data & Cloud: Python, SQL, Google Cloud Platform (GCP/Vertex AI), AWS, bases de données volumineuses

Outils & Déploiement: Dataiku, Docker, Git, architectures distribuées, industrialisation de modèles

Experience

AI Engineer / Data Scientist Jan 2026 – Present
Kairos Research AI *Paris, France*

- **Quantitative Modeling & Elo**: Engineered custom Elo and bivariate Poisson rating systems for soccer models, optimizing probability calibration and generating high-alpha market signals.
- **Player Props Predictive Models**: Built end-to-end predictive pipelines for basketball player props and sequential player performance using LightGBM and deep learning architectures.
- **Portfolio & Execution Strategy**: Implemented de-vigging algorithms and Bayesian portfolio-sizing modules (Kelly Criterion) to optimize risk management and dynamic capital allocation.
- **Data Engineering & Backtesting**: Developed scalable data pipelines and backtesting frameworks to process multi-sport telemetry data, streamline model validation, and speed up iteration cycles.

AI Engineer Dec 2024 – Dec 2025
Libertify *Paris, France*

- Built production RAG pipelines and optimized embedding retrieval systems for cross-language semantic search; developed prompt-based generative agents (LLMs) deployed on Vertex AI and AWS Lambda.
- Developed a multimodal image retrieval system and deployed text-to-image diffusion pipelines (Imagen/Vertex AI) for automated video background generation at scale.

AI Engineer Intern May 2024 – Oct 2024
Renault *Guyancourt, France*

- Fine-tuned transformer models for document understanding; applied YOLO layout detection to segment heterogeneous scanned PDFs for automated structural comparison across procurement responses.
- Developed interpretable fault-detection models using XGBoost with SHAP feature attribution, applying SMOTE for class imbalance in predictive maintenance pipelines.

Projects

Time Series Anomaly Detection (Genvia) Oct 2023 – Apr 2024
Predictive Maintenance with Domain Shift *Python, VAE, Domain Adaptation*

- Designed domain-adaptive VAEs for industrial fault detection (custom encoder/decoder, KL-weighted loss) and GASF/GADF transformations, achieving 90% accuracy across domain-shifted sensor data (*Prix Dzung Tran*).
- Co-authored *"Interpretable Domain-Adaptive Deep Learning for Noisy Time-Series Failure Prediction"*.

Education

CentraleSupélec Gif-sur-Yvette, FR
Engineering Degree (M.S. equiv.) in Artificial Intelligence *Aug 2021 – Dec 2024*

Preparatory Classes (CPGE) Rabat, MA
MPSI & MP – Math, Physics, Engineering* *Sep 2019 – Jun 2021*

Languages

Arabic: Native | **French:** Bilingual Proficiency | **English:** Full Professional Proficiency